

Aathavan Nithiyananthan

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A passionate Engineering undergraduate who aspires to learn, grow, and network with like-minded individuals to turn innovative ideas into reality.

Education

- **University of Moratuwa** Moratuwa, Sri Lanka
Bachelor of Science (Honours) in Engineering Undergraduate Mar 2021 - Present
Specializing in: *Computer Science and Engineering*
Stream: *Data Science and Engineering*
Current CGPA: 3.70
- **Jaffna Hindu College** Jaffna, Sri Lanka
GCE Advanced Level Jan 2017 - Aug 2019
Subjects: *Distinction in Physics, Chemistry, Combined mathematics and General English*
Z-score 2.6381

Experience

- **University of Moratuwa** Teaching Assistant
Worked part-time as Teaching Assistance Jun 2024 - Oct 2024
 - **Teaching Assistant:** For CS3051 Data Science and Engineering Project module.
- **London Stock Exchange Groups** Software Engineering Intern
Worked as Software Engineering Intern for Industrial Training Nov 2023 - May 2024
 - **TRADEcho:** Worked in a team which is focused on improving the non-functional requirements of the MiFID II reporting services application TRADEcho.
 - **Research Project:** Worked in a project to develop Machine Learning based methods to identify software vulnerabilities.
- **University of Jaffna** Research Assistant
Worked part-time as a research assistant Oct 2023 - Jan 2023
 - **Project objective:** To build a Tamil conversational bot using LLMs.
 - **Research:** Fine-tuned GPT-3.5 to respond to e-commerce queries in Tamil with an accuracy of 97.3%.
- **Yarl Salesforce Ohana** Volunteer
Hackforce 22, YarlDreamin'23 organizing committee member May 2022 - Sep 2023
 - **Event coordination:** Worked as lead for technical works of YarlDreamin'23
 - **Mentoring:** Mentored teams for Hackforce'22 to develop their salesforce platform
- **Yarl It Hub** Volunteer
Yarl Geek Challenge Junior Mentor Mar 2022 - Aug 2022
 - **Workshops:** Conducted workshops on basic web development and hardware development. Introduced HTML, CSS, JavaScript and Arduino platform to students.
 - **Mentoring:** Supported to teams to improve their web and hardware solutions.
- **Tamil Wikipedia** Volunteer
Contributor Mar 2013 - Present
 - **Technical tools:** Adapted technical tools such as Save&Edit to function seamlessly on Tamil Wikipedia
 - **Article creation:** Have created 130+ Articles
 - **Edits:** Have done 10000+ Edits to Tamil Wikipedia

Research Projects

- **Final Year Research Project (Ongoing):** Research project titled "Metaphor Surface Realization and Metaphor Sentence Generation for Tamil Language". Developing a system leveraging large language models to generate contextually relevant and linguistically nuanced simile sentences.
- **Science Research Projects Competition:** Research project titled "Study of Adolescents' Introversion and Extraversion Traits and the effect of these traits in Social Media Usage" for Science Research Projects Competition by National Science Foundation.

Academic Projects

- **5G Energy Consumption Modeling - Group:** Modeling and predicting energy usage for 5G network stations using datasets of configurations. Tasks involve creating a simple application and dashboard to run our model and visualize the findings.
- **Human Resource Management System - Group:** Developed HR Management System streamlining PIM and Absence tasks. Built on React, Node.js, MySQL with customizable attributes and dynamic leave management.
- **Email Client - Individual:** The Email Client project is a Java-based desktop application that leverages Object-Oriented Programming (OOP) principles and the Simple Mail Transfer Protocol (SMTP) through the javax.mail API.
- **Nano Processor Design in Vivado - Individual:** The Nano Processor project involves designing and implementing a 4-bit microprocessor capable of executing a simple set of instructions.

Hackathon Projects

Projects I participated in as a team member for various hackathons, all of which were led by me.

- **Sportswise (2023):** A sports team management and player performance analytics system.
 - **Tech:** Salesforce, Salesforce Aura components, Thingspeak
- **ProSeeds (2021) :** A competition management system which can be used by organizations to organize and host competitions with less man power.
 - **Tech:** Salesforce, Salesforce community cloud, Salesforce lightning web components
- **Flowsulin (2021):** Flowsulin is a virtual assistance to help people with diabetes.
 - **Tech:** Voiceflow API, JavaScript
- **Varaipadavi (2017):** Varaipadavi is a Web Application that helps students to develop their map marking and remembering skill in a new fun way.
 - **Tech:** HTML, CSS, Bootstrap, JavaScript, Google Sheets
- **JDIMan (2016):** A mobile and web application to rescue people from natural disasters and to organize relief activities.
 - **Tech:** HTML, CSS, JavaScript, Google Maps API
- **RememberMe (2015):** A mobile application that is created on ionic platform to effectively remember works to do.
 - **Tech:** Ionic, JavaScript, MySQL

Honors and Awards

- Winners of Hackforce Seniors 2023 - *Project Sportswise*
- University Colours award for Basketball - Mar, 2022
- 2nd Runners up of Hackforce'21 & special award for most innovative idea - *Project ProSeeds*
- Voice flow API prize finalist of Canada's biggest hackathon Hack the North 2021 - *Project Flowsulin*
- Finalists of solution hackathon - Climathon by Climate KIC supported by EIT - Nov, 2020
- Sri Lanka Science & Engineering Fair 2018 Finalist - May, 2018
- Science Research Projects Competition 2017 Winner - May 2018
- Yarl Geek Challenge Season 6 - Best web application, 2017 - *Project Varaipadavi*

Extracurricular Activities

- Basketball Player - **Captain** of University of Moratuwa Basketball Team for the year 2024.
- Classical Dancer - Performed Bharatham and Kandyam Dance in various university events.